Aryan Thakar

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571-315-6939 | Washington D.C. Metro Area (Willing to Relocate) | U.S. Citizen; Eligible for Security Clearance

Education

University of Maryland — College Park, MD
Bachelor of Science in Computer Science; Minor in Business — *Expected Dec 2026*
- Dean's List, Spring 2024
- Relevant Coursework: Data Structures & Algorithms, Computer Systems (C, OS concepts), Artificial Intelligence, Machine Learning, Data Science, Web Application Development, Discrete Structures

Technical Skills

Languages: Java, Python, C, Rust, OCaml, JavaScript, TypeScript, SQL, R, HTML/CSS
Frameworks/Tools: React, Node.js, FastAPI, Flask, JUnit, Auth0, Docker, Celery, Redis, REST API, Vercel, NoSQL
AI/ML: TensorFlow, PyTorch, OpenCV, MediaPipe, dlib, Fairlearn, LIME, Matplotlib, NLP, GenAI, LangGraph

Experience

AI Security Intern — May 2026 – Aug 2026
The MITRE Corporation — *McLean, VA*
- Built an experimental pipeline for testing robustness of Open Policy Agents using attack cases generated with Promptfoo, **reducing successful attacks by 30%** for output-rail and tool-authentication components.
- Increased throughput and usability of AI vulnerability analysis software by developing a Docker-containerized React/FastAPI web application using Redis for batch queueing and Celery workers for concurrent processing; identified parser-validation vulnerabilities through testing varied system prompts.

Course Facilitator for CMSC398N: Ethics in the Age of AI — Aug 2026 – Dec 2026
University of Maryland — *College Park, MD*
- Co-facilitate a student-led course on ethical issues in computer science and AI, guiding discussions of real-world impacts and responsible decision-making.

Course Facilitator for CMSC388J: Building Secure Web Applications — Aug 2025 – Dec 2025
University of Maryland — *College Park, MD*
- Co-facilitated course lectures covering secure full-stack development using Flask, Django, MongoDB, and Svelte/React & collaboratively **modernized curriculum** by adding scalability, ETL, proxies, and caching concepts.

Campus Ambassador for Perplexity — Sep 2025 – Jan 2026
Perplexity AI — *Remote*
- Partnered with multiple student organizations to design and lead technical workshops covering Comet's MCP capabilities for **30+ students** and generating **300+ resource clicks**

Creative Software Developer Intern — May 2025 – Aug 2025
RedPeg Marketing — *Alexandria, VA*
- Improved RAG chatbot response quality by 30% through pipeline adjustments that increased answer relevance, reduced latency, and strengthened brand-voice adherence.
- Automated calendar-event organization with a CustomGPT workflow using Microsoft Graph API and OpenAI function calling, saving an estimated 30–40 minutes per employee weekly.

Research Fellow — Jun 2024 – Jan 2025
NSF Institute for Trustworthy AI in Law and Society (TRAILS) — *College Park, MD*
- Designed inclusive interview protocols to reduce bias in automated video interviewing systems.
- Implemented feature extraction methods (dlib, OpenCV, face-utils) improving accuracy by **0.1 MSE**.

Other Projects

DocuWell | OpenAI, React, MongoDB, Auth0 — Sep 2024
- Fine-tuned OpenAI chatbot for patient–doctor communication; boosted interaction quality by **20%** and enabled more accurate root-cause diagnosis (View code).

Identifying Areas of Bias in Datasets | Python, Fairlearn, LIME — Aug 2022 – May 2023
- Conducted a mini meta-analysis on U.S. Census + UCI Heart Disease datasets, uncovering gender over-weighting as a source of misclassification (View code).

Authentic Screen Talk | Flask, Jinja, SQLAlchemy — Jul 2025 - Aug 2025
- Built full-stack movie/TV review site prompting users to share thoughtful reviews beyond ratings via intelligently generated prompts; built with Flask, Jinja templates, and SQLAlchemy (View code).

Certifications

Foundational AI Engineering | Applied AI Engineering | Technical AI Safety